

Introduction to



Firedrake

Computational Fluid Dynamics

A.Y. 2025/2026

Ivan Fumagalli



POLITECNICO
MILANO 1863

DIPARTIMENTO
DI MATEMATICA

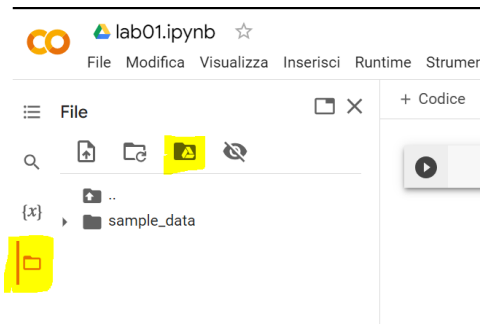


Firedrake on Colab



Requirements: a browser and a Google account

- 1 Go to <https://colab.research.google.com> and create a new notebook.
- 2 Mount your Google Drive space (pop-up window could appear; after, you may need to **reload** the notebook).



Firedrake on Colab



Requirements: a browser and a Google account

- 1 Go to <https://colab.research.google.com> and create a new notebook.
- 2 Mount your Google Drive space (pop-up window could appear; after, you may need to **reload** the notebook).
- 3 Install **Firedrake** as in <https://fem-on-colab.github.io/packages.html> (**takes a while ...**; you may need to run the cell **twice**).

Firedrake on Colab



Requirements: a browser and a Google account

- 1 Go to <https://colab.research.google.com> and create a new notebook.
- 2 Mount your Google Drive space (pop-up window could appear; after, you may need to **reload** the notebook).
- 3 Install **Firedrake** as in <https://fem-on-colab.github.io/packages.html> **takes a while ...**; you may need to run the cell **twice**).
- 4 Have you obtained this?
You are ready to go!

```
#####  
# This installation is offered by FEM on Colab, an open-source project #  
# developed and maintained at Università Cattolica del Sacro Cuore #  
# by Dr. Francesco Ballarin. Please see https://fem-on-colab.github.io/ #  
# for more details, including a list of further available packages #  
# and how to sponsor the development or contribute to the project. #  
#  
# We are conducting an informal survey on FEM on Colab usage by our users. #  
# The survey is anonymous, and its compilation will typically only require #  
# a couple of minutes of your time. If you wish, give us your feedback at #  
# https://forms.gle/36sZ2H0vPouu8Xh7 #  
#####
```

Firedrake on Colab



Requirements: a browser and a Google account

- 1 Go to <https://colab.research.google.com> and create a new notebook.
- 2 Mount your Google Drive space (pop-up window could appear; after, you may need to **reload** the notebook).
- 3 Install **Firedrake** as in <https://fem-on-colab.github.io/packages.html> **takes a while ...**; you may need to run the cell **twice**).
- 4 Have you obtained this?
You are ready to go!

```
#####  
# This installation is offered by FEM on Colab, an open-source project #  
# developed and maintained at Università Cattolica del Sacro Cuore #  
# by Dr. Francesco Ballarín. Please see https://fem-on-colab.github.io/ #  
# for more details, including a list of further available packages #  
# and how to sponsor the development or contribute to the project. #  
#  
# We are conducting an informal survey on FEM on Colab usage by our users. #  
# The survey is anonymous, and its compilation will typically only require #  
# a couple of minutes of your time. If you wish, give us your feedback at #  
# https://forms.gle/36s72H0vPouu8Xh7 #  
#####
```

```
from firedrake import *  
import numpy as np  
import matplotlib.pyplot as plt
```

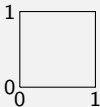
Steps of FEM implementation

- ① mesh generation
- ② creation of finite element spaces
- ③ assembling matrices and vectors
- ④ solving the algebraic problem
- ⑤ save or visualize the solution

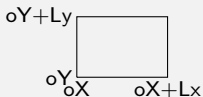
Mesh generation

Rectangular domains

```
mesh = UnitSquareMesh(nx, ny,  
    diagonal='left',  
    quadrilateral=False)
```



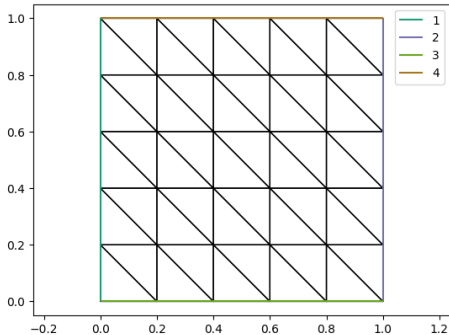
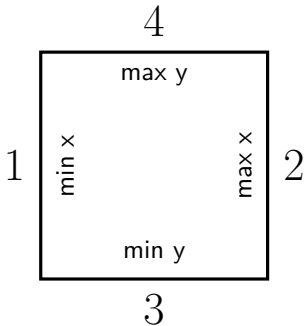
```
mesh = RectangleMesh(nx, ny,  
    Lx, Ly,  
    originX=0.0, originY=0.0,  
    diagonal='left',  
    quadrilateral=False)
```



nx, ny : subdivisions along x and y axis

Mesh generation

Boundary labels



```
fig, ax = plt.subplots()  
triplot(mesh, axes=ax)  
ax.legend()
```


Finite element spaces

```
V = FunctionSpace(mesh, family,  
degree)
```

family = 'P'/'CG'/Lagrange' (continuous FE),
'DG'/'Discontinuous Lagrange', ...
(full list [here](#))

degree = 0 (NOT for 'CG'), 1, 2, ...

N.B. We use the FunctionSpace by Firedrake (documented [here](#)),
not the one in UFL (see [here](#)).

The diffusion-reaction problem

Weak formulation

$$\begin{cases} -\Delta u + u = f, & \text{in } \Omega := (0, 1)^2, \\ \nabla u \cdot \mathbf{n} = 0, & \text{on } \partial\Omega. \end{cases}$$

The diffusion-reaction problem

Weak formulation

$$\begin{cases} -\Delta u + u = f, & \text{in } \Omega := (0, 1)^2, \\ \nabla u \cdot \mathbf{n} = 0, & \text{on } \partial\Omega. \end{cases}$$

Find $u \in V := H^1(\Omega)$ such that

$$\int_{\Omega} (\nabla u \cdot \nabla v + uv) \, d\mathbf{x} = \int_{\Omega} fv \, d\mathbf{x} \quad \forall v \in V.$$

The diffusion-reaction problem

Galerkin projection - Finite Element Method

mesh $\mathcal{T}_h$ of Ω

$$X_h^p = \{u \in C^0(\overline{\Omega}) : u|_K \in \mathbb{P}^p(K) \quad \forall K \in \mathcal{T}_h\}$$

$$V_h = V \cap X_h^1$$

Find $u_h \in V_h$ such that

$$\int_{\Omega} (\nabla u_h \cdot \nabla v_h + u_h v_h) \, d\mathbf{x} = \int_{\Omega} f v_h \, d\mathbf{x} \quad \forall v_h \in V_h$$

The diffusion-reaction problem

Firedrake code

```
u = TrialFunction(V) # placeholders
v = TestFunction(V) # not func's

a = dot(grad(u), grad(v)) * dx
    + u * v * dx
L = f * v * dx

u_h = Function(V) # actual function
solve(a == L, u_h)
```

The diffusion-reaction problem

Non-homogeneous Neumann BC - Weak formulation

$$\begin{cases} -\Delta u + u = 0, & \text{in } \Omega := (0, 1)^2, \\ \nabla u \cdot \mathbf{n} = d, & \text{on } \partial\Omega. \end{cases}$$

The diffusion-reaction problem

Non-homogeneous Neumann BC - Weak formulation

$$\begin{cases} -\Delta u + u = 0, & \text{in } \Omega := (0, 1)^2, \\ \nabla u \cdot \mathbf{n} = d, & \text{on } \partial\Omega. \end{cases}$$

Find $u \in V := H^1(\Omega)$ such that

$$\int_{\Omega} \nabla u \cdot \nabla v + uv \, d\mathbf{x} = \int_{\partial\Omega} d v \, ds \quad \forall v \in V.$$

The diffusion-reaction problem

Non-homogeneous Neumann BC - Firedrake code

```
a = dot(grad(u), grad(v)) * dx
    + u * v * dx
L = d * v * ds
# L = d*v * ds(1) + d*v * ds(2)
#     + d*v * ds(3) + d*v * ds(4)

u_h = Function(V)
solve(a == L, u_h)
```


Visualization

matplotlib

```
fig, ax = plt.subplots()  
q = tripcolor(u_h, axes=ax)  
fig.colorbar(q) # show the colorbar
```

ParaView

Software for enhanced visualization
(later in the course)

<https://www.paraview.org/download>

Algebraic formulation of FEM

Let $V_h \subset V$ be a finite element space with dimension N_h and $\{\varphi_i\}_{i=1}^{N_h}$ its basis. Then, the FE approximation of the problem is:

Find $u_h = \sum_{j=1}^{N_h} u_j \varphi_j \in V_h$ such that

$$\int_{\Omega} (\nabla u_h \cdot \nabla \varphi_i + u_h \varphi_i) d\mathbf{x} = \int_{\partial\Omega} d \varphi_i ds$$
$$\forall i = 0, \dots, N_h$$

Algebraic formulation of FEM

$$\int_{\Omega} \sum_{j=0}^{N_h} u_j (\nabla \varphi_j \cdot \nabla \varphi_i + \varphi_j \varphi_i) d\mathbf{x} = \int_{\partial\Omega} d\varphi_i ds$$

$$\forall i = 0, \dots, N_h$$

$$\Updownarrow$$

$$A\mathbf{U} = \mathbf{b} \text{ with}$$

$$A_{ij} = \int_{\Omega} (\nabla \varphi_j \cdot \nabla \varphi_i + \varphi_j \varphi_i) d\mathbf{x},$$

$$b_i = \int_{\partial\Omega} d\varphi_i ds$$

Algebraic formulation of FEM

Firedrake code

```
A = assemble(a)
rhs = assemble(L)

solver = LinearSolver(A,
                      solver_parameters={...})

u_h = Function(V)
solver.solve(u_h, rhs)
```

The diffusion-reaction problem

Dirichlet boundary conditions

$$\begin{cases} -\Delta u + u = 0, & \text{in } \Omega := (0, 1)^2, \\ u = g, & \text{on } \partial\Omega. \end{cases}$$

The diffusion-reaction problem

Dirichlet boundary conditions

$$\begin{cases} -\Delta u + u = 0, & \text{in } \Omega := (0, 1)^2, \\ u = g, & \text{on } \partial\Omega. \end{cases}$$

$$V_0 = H_0^1(\Omega), \quad V_D = \{v \in H^1(\Omega) : v|_{\partial\Omega} = g\},$$

lifting operator $\mathcal{R} : H^{1/2}(\partial\Omega) \rightarrow V_D$

The diffusion-reaction problem

Dirichlet boundary conditions

$$\begin{cases} -\Delta u + u = 0, & \text{in } \Omega := (0, 1)^2, \\ u = g, & \text{on } \partial\Omega. \end{cases}$$

$$V_0 = H_0^1(\Omega), \quad V_D = \{v \in H^1(\Omega) : v|_{\partial\Omega} = g\},$$

lifting operator $\mathcal{R} : H^{1/2}(\partial\Omega) \rightarrow V_D$

Find $u = \check{u} + \mathcal{R}g \in V_D$ with $\check{u} \in V_0$ s.t.

$$a(\check{u}, v) = F(v) - a(\mathcal{R}g, v) \quad \forall v \in V_0$$

The diffusion-reaction problem

Dirichlet boundary conditions - Firedrake code

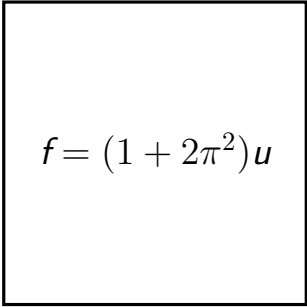
```
x = SpatialCoordinate(mesh)
g = x[0] * sin(x[1]) # example

bc = DirichletBC(V, g, [1,2,3,4])
bcs = list(bc)
solve(a == L, u_h, bcs=bcs,
      solver_parameters={...})
```


Exercise

$$-\Delta u + u = f \quad \text{in } \Omega = (0, 1)^2$$

$$\nabla u \cdot \mathbf{n} = 0$$


$$u = 0 \quad f = (1 + 2\pi^2)u \quad u = 0$$

$$\nabla u \cdot \mathbf{n} = 0$$

$$u_{\text{exact}} = \sin(\pi x) \cos(\pi y)$$